

Counting Reorders

Time limit: 1.00 s

Memory limit: 512 MB

Calculate the number of ways you can reorder the characters of a string so that no two adjacent characters are the same.

For example, the answer for aabc is 6, because the possible orders are abac, abca, acab, acba, baca, and caba.

Input

The only input line has a string that consists of n characters between a–z.

Output

Print an integer: the answer modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 5000$

Example

Input	Output
aabc	6